Kun Huang

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Academic Appointments

Texas A&M University | Visiting Assistant Professor, July 2026-present

Texas A&M University | Postdoctoral Research Associate, September 2023-July 2026. Advisors: Huiyan Sang and Xiyu Peng

Education

Tsinghua University | Ph.D. in Statistics, 2018-2023. Advisor: Lijian Yang

Tsinghua University | B.E. in Mechanical Engineering, minor in Statistics, 2014-2018

Research Interests

Bayesian spatial machine learning; learning for heterogeneous and structured data; high-dimensional statistics; spatial causal inference; functional data analysis; spatio-temporal point processes. Applications in spatial omics, transportation, environmental science, biomedical sciences, and AI-enabled systems.

Peer-Reviewed Publications - Methodology and Theory

1. **Huang, K.** and Sang, H. (2026). Consistent Bayesian spatial domain partitioning using predictive spanning tree methods. Annals of Statistics, 54, 2432-2454. **NeurIPS 2026 Journal Track.**
2. **Huang, K.**, Chen, X., Guan, Y. and Li, Y. (2025). Partially linear single-index models and functional principal component analysis of spatially and temporally indexed point processes. Journal of Business & Economic Statistics, 43, 1077-1091.
3. Zheng, S., **Huang, K.** and Yang, L. (2025). Inference for dependent error functional data: covariance function. Electronic Journal of Statistics, 19, 5216-5248.
4. Chen, X., **Huang, K.**, Wu, W. and Jiang, H. (2025). Detecting multiple change points by exploiting their spatio-temporal correlations: A Bayesian hierarchical approach. INFORMS Journal on Data Science, 4, 133-153.
5. **Huang, K.**, Zheng, S. and Yang, L. (2022). Inference for dependent error functional data with application to event related potentials. TEST, 31, 1100-1120.
6. Chen, X., **Huang, K.** and Jiang, H. (2022). Detecting changes in the spatiotemporal pattern of bike sharing: a change-point topic model. IEEE Transactions on Intelligent Transportation Systems, 23, 18361-18377.

Peer-Reviewed Publications - Interdisciplinary Applications

7. Shi, S., Chen, W., Jiang, Y., Chen, K., Liao, Y. and **Huang, K.** (2023). A more accurate method to estimate muscle mass: a new estimation equation. Journal of Cachexia, Sarcopenia and Muscle, 14, 1753-1761.
8. Chen, W., Shi, S., Jiang, Y., Chen, K., Liao, Y., Huang, R. and **Huang, K.** (2022). Association of sarcopenia with ideal cardiovascular health metrics among US adults. BMJ Open, 12, e061789.
9. Shi, S., Jiang, Y., Chen, W., Chen, K., Liao, Y. and **Huang, K.** (2022). Diagnostic and prognostic value of the Creatinine/Cystatin C ratio for low muscle mass evaluation among US adults. Frontiers in Nutrition, 9.
10. Chen, W., Shi, S., Jiang, Y., Chen, L., Liao, Y., Chen, K. and **Huang, K.** (2022). Association between riboflavin intake and telomere length: a cross-sectional study from NHANES 1999-2002. Frontiers in Nutrition, 744397.
11. Hu, C., Zhu, K., **Huang, K.**, Yu, B., Jiang, W., Peng, K. and Wang, F. (2022). Using natural intervention to promote subjective well-being of essential workers during public-health crises. Journal of Environmental Psychology, 79, 101745.

12. **Huang, K.**, Chen, D., Wang, F. and Yang, L. (2021). Prediction of dispositional dialectical thinking from resting-state electroencephalography. Brain and Behavior, 11, e2327.
13. Zhang, Y., Wang, C., Wu, F., **Huang, K.**, Yang, L. and Ji, L. (2020). Prediction of working memory ability based on EEG by functional data analysis. Journal of Neuroscience Methods, 333, 108552.

Current Manuscripts

1. **Huang, K.**, Peng, X., Sang, H. and Lu, L. Consistent Bayesian local spatial feature selection with application to spatial multimodal omics. Major revision at Journal of the American Statistical Association. arXiv:2605.30658.
2. **Huang, K.**, Peng, X. and Sang, H. A consistent Bayesian adaptive quotient graph framework for graph-informed random partition models. Manuscript.
3. **Huang, K.**, Yang, S., Li, Y. and Guan, Y. Interference-aware causal estimation for marked spatio-temporal point processes: assessing the effects of California wildfires on PM2.5. Manuscript under review.
4. Ghosh, A., Sang, H. and **Huang, K.** Consistent spatial clustering: a general framework. Working paper.

Selected Awards and Presentations

NSF Travel Award, ICSA Applied Statistics Symposium (2025); First Class Scholarship for Comprehensive Performance, Tsinghua University (2022); Peking-Tsinghua Joint Statistics Colloquium Excellent Poster Award (2021).

2026 Joint Statistical Meetings (Boston); 2025 ICSA Applied Statistics Symposium (Storrs); 2024 Joint Statistical Meetings (Portland).

Teaching Experience

Instructor: STAT 302, Statistical Methods, Texas A&M University (Fall 2024; Fall 2026).

Teaching Assistant: Computational Statistics, Tsinghua University (Fall 2018; Fall 2019); course-material preparation and teaching committee support (Spring 2019).

Academic Service

Referee for Journal of the American Statistical Association, Journal of Computational and Graphical Statistics, Journal of Multivariate Analysis, Statistica Sinica, Stat, and other journals.